QSETH-conditional quantum lower bounds for high-dimensional closest pair and related geometric search

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Editorial scope. One conditional quantum fine-grained hardness mechanism with full SAT/OV/BCP/CP and related integer-OV reduction family. Related dimensions/exponents and bichromatic consequences grouped once. QSETH is assumed;this task does not assert an unconditional separation or duplicate constant-dimensional algorithm GA101.

Allowed standard prerequisites (editorial).

- QSETH,Grover/minimum finding,known contact-dimension realization/covering and integer-OV encodings. QSETH remains an explicit assumption,not a theorem. Formal Theorem31/Lemma33[29] and Theorem37[17] retained as cited prerequisite constructions with full stated hypotheses;all author's quantum-oracle realization,cost/probability analysis and related reductions are retained. Full52 original bibliography entries retained.

2 Preliminaries

Definition 6 (Distance measure)**.** *For any two vectors $a, b \in \mathbb{R}^d$, the distance between them in the ℓ_2-metric is denoted by $\|a-b\| = \left(\sum_{i=1}^{d} |a_i - b_i|^2\right)^{1/2}$. Their distance in the ℓ_0-metric (Hamming distance) is denoted by $\|a-b\|_0 = |\{i \in [d] : a_i \neq b_i\}|$, i.e., the number of coordinates on which a and b differ.*

2.1 Quantum query model

We consider the quantum query model in this work. Let $X := \{x_1, \ldots, x_n\}$ be a set of n input points and $\mathcal{O}_X$ be the corresponding oracle. We can access the i-th data point x_i by making the query

$$|i\rangle |0\rangle \xrightarrow{\mathcal{O}_X} |i\rangle |x_i\rangle , \tag{1}$$

and we can make queries to elements in X in superposition. Note that $\mathcal{O}_X$ is an unitary transformation in the formula above. Hence, a quantum algorithm with access to $\mathcal{O}_X$ can be represented as a sequence of unitary transformations.

Consider a quantum algorithm $\mathcal{A}$ with access to an oracle $\mathcal{O}$ and a initial state $|0\rangle := |0\rangle_Q |0\rangle_A |0\rangle_W$, where the registers Q and A are for the queries and the answers from the oracle, and the register W is the working space which is always hold by $\mathcal{A}$. Then, we can represent the algorithm as

$$U_T \mathcal{O} U_{T-1} \cdots \mathcal{O} U_1 |0\rangle . \tag{2}$$

Let $|\psi\rangle_i = U_i \mathcal{O} \cdots \mathcal{O} U_1 |0\rangle := \sum_{i,z} |i\rangle_Q |0\rangle_A |z\rangle_W$ be the state right before applying the i-th $\mathcal{O}$, then

$$\mathcal{O} |\psi\rangle_i := \sum_{i,z} |i\rangle_Q |x_i\rangle_A |z\rangle_W . \tag{3}$$

2.2 Quantum subroutine for unstructured searching and minimum finding

Definition 7 (Unstructured search)**.** *Given a set P of n elements in $\{0,1\}$, decide whether there exists a 1 in P.*

Theorem 8 (Grover's search algorithm [24, 37])**.** *There is a quantum algorithm for unstructured search with running time $O(\sqrt{n})$.*

By Theorem 8 and BBBV's argument [9], the quantum time complexity of unstructured search is $\Theta(\sqrt{n})$. We can also get a $\widetilde{O}(\sqrt{n})$ quantum algorithm for minimum finding by combining Grover's search algorithm and binary search.

Theorem 9 (Quantum minimum finding [23])**.** *There is a quantum algorithm that finds from a set of n elements with values in $\mathbb{R}$, the index of the minimum element of the set, with success probability $\frac{1}{2}$ and run time $\widetilde{O}(\sqrt{n})$.*

2.3 Problem definitions

In this subsection, we first formally define OV, CP, and BCP. Then we show the folklore algorithms for CP, BCP, and OV by Grover's algorithm, which run in time $\widetilde{O}(n)$.

Definition 10 (Orthogonal Vectors, OV)**.** *Given two sets A, B of n vectors in $\{0,1\}^d$ as input, find a pair of vectors $a \in A$, $b \in B$ such that $\langle a, b\rangle = 0$, where the inner product is taken in $\mathbb{Z}$.*[1]

We denote OV with input length n and dimension d as $\mathsf{OV}_{n,d}$. We will use this notation when we need to specify the parameters in the following sections.

Definition 11 (Closest Pair Problem, CP)**.** *Given a set P of n points in $\mathbb{R}^d$ and a distance measure Δ, find a pair of distinct points $a, b \in P$ such that $\Delta(a, b)$ is the smallest among all distinct pairs in P.*

Similar to OV, we denote CP with input length n and dimension d as $\mathsf{CP}_{n,d}$. We will use this notation when the parameters in the following sections are required to be specified. Note that in this work, we consider $\Delta(a, b) = \|a - b\|$ as the distance measure for CP and BCP.

[1] Our definition is slightly different than some of the literature, for example, [18], which is searching among pairs inside one set. Those two definitions are equivalent up to constant in complexities.

Definition 12 (Bichromatic Closest Pair Problem, BCP)**.** *Given two sets A, B of n points in $\mathbb{R}^d$ and a distance measure Δ, find a pair of points $a \in A$, $b \in B$ such that*

$$\Delta(a, b) = \min_{a \in A, b \in B} \Delta(a, b). \tag{4}$$

We also define an approximate version of BCP as follows.

Definition 13 ($(1+\xi)$-approximate Bichromatic Closest Pair Problem, $(1+\xi)$-BCP)**.** *Given two sets A, B of n points $\in \mathbb{R}^d$ and a distance measure Δ, find a pair of points $a \in A$, $b \in B$ such that*

$$\Delta(a, b) \leq (1 + \xi) \min_{a \in A, b \in B} \Delta(a, b). \tag{5}$$

Same as CP, we use $\mathsf{BCP}_{n,d}$ and $(1+\xi)$-$\mathsf{BCP}_{n,d}$ to specify the parameters.

Definition 14 (Element Distinctness Problem, ED)**.** *Let $f : [n] \to [m]$ be a given function. Decide whether there exist distinct $i, j \in [n]$ such that $f(i) = f(j)$.*

For this problem, Ambainis [5] gave a quantum algorithm with time complexity $\widetilde{O}(n^{2/3})$, which matches the lower bound proved by Aaronson and Shi [1] up to a polylogarithmic factor.

Theorem 15. *There are $\widetilde{O}(n)$-time quantum algorithms for CP and BCP when $d = O(\operatorname{poly} \log n)$.*

Proof. We can solve CP and BCP by searching the minimum distance through all pairs by the algorithm of Theorem 9. There are $O(n^2)$ pairs and checking each pair took $O(d)$ time, so the total running time is $O(nd)$. For $d = O(\operatorname{poly} \log n)$, the time complexity equals to $\widetilde{O}(n)$. □

2.4 Fine-grained complexity

As we have mentioned earlier in the introduction, a fine-grained reduction from problem P to Q with conjectured lower bounds $p(n)$ and $q(n)$, respectively, has the property that if we can improve the $q(n)$ time for Q, then we can also improve the $p(n)$ time for P. We give the formal definition by Williams [46] in below.

Definition 16 (Fine-grained reduction, [46])**.** *Let $p(n)$ and $q(n)$ be non-decreasing functions of n. Problem P is (p, q)-reducible to problem Q, denoted as $(\mathsf{P}, p) \leq_{\mathrm{FG}} (\mathsf{Q}, q)$, if for every ϵ, there exist $\delta > 0$, an algorithm R for solving P with access to an oracle for Q, a constant d, and an integer $k(n)$, such that for every $n \geq 1$, the algorithm R takes any instance of P of size n and*

- *R runs in at most $d \cdot (p(n))^{1-\delta}$-time,*
- *R produces at most $k(n)$ instances of Q adaptively, that is, the jth instance X_j is a function of $\{(X_i, y_i)\}_{1 \leq i < j}$ where X_i is the ith instance produced and y_i is the answer of the oracle for Q on instance X_i, and*

- *the sizes n_i of the instances X_i for any choice of oracle answers y_i obeys the inequality*

$$\sum_{i=1}^{k(n)} (q(n_i))^{1-\epsilon} \leq d \cdot (p(n))^{1-\delta}. \tag{6}$$

Let $(\mathsf{P}, p) \leq_{\mathrm{FG}} (\mathsf{Q}, q)$ for some non-decreasing function $p(n)$ and $q(n)$. If for every $\epsilon > 0$, we can solve problem Q in time $q(n)^{1-\epsilon}$ with probability 1 for all input length n, then there exists a $\delta > 0$ such that we can solve the problem P in time $p(n)^{1-\delta}$ by Eq. (6).

Here are some known results about fine-grained reductions.

Theorem 17 ([29, 49])**.**

$$(\mathsf{CNF\text{-}SAT}_n, 2^n) \leq_{\mathrm{FG}} (\mathsf{OV}_{n_1,d_1}, n_1^2) \leq_{\mathrm{FG}} (\mathsf{BCP}_{n_2,d_2}, n_2^2) \leq_{\mathrm{FG}} (\mathsf{CP}_{n_3,d_3}, n_3^2), \tag{7}$$

where $d_1 = \Theta(\log n_1)$, $d_2 = \Theta(\log n_2)$ and $d_3 = (\log n_3)^{\Omega(1)}$.

Remark 18. *The second reduction from OV to BCP has been improved to $d_2 = 2^{O(\log^* n)}$ by Chen [17].*

There are several plausible hypotheses in fine-grained complexity, which can imply conditional hardness results for many interesting problems. We first give the definition of the strong exponential time hypothesis (SETH).

Hypothesis 19 (Strong Exponential Time Hypothesis, SETH)**.** *For every $\epsilon > 0$, there exists a $k = k(\epsilon) \in \mathbb{N}$ such that no algorithm can solve k-SAT (i.e., satisfiability on a CNF of width k) in $O(2^{(1-\epsilon)m})$ time where m is the number of variables. Moreover, this holds even when the number of clauses is at most $c(\epsilon) \cdot m$ where $c(\epsilon)$ denotes a constant that depends only on ϵ.*

Another popular conjecture is the orthogonal vector hypothesis (OVH):

Definition 20 (Orthogonal Vector Hypothesis, OVH)**.** *For every $\epsilon > 0$, there exists a $c \geq 1$ such that $\mathsf{OV}_{n,d}$ requires $n^{2-\epsilon}$ time when $d = c \log n$.*

Remark 21. *Under SETH, we can have the following conclusions from Theorem 17:*

- OVH *is true.*
- *For all $\epsilon > 0$, there exists a $c > 0$ such that $\mathsf{BCP}_{n,c\log n}$ cannot be solved by any randomized algorithm in time $O(n^{2-\epsilon})$.*
- *For all $\epsilon > 0$, there exists a $c > 0$ such that $\mathsf{CP}_{n,(\log n)^c}$ cannot be solved by any randomized algorithm in time $O(n^{2-\epsilon})$.*

2.5 The framework for quantum walk search

In this subsection, we review the quantum walk framework for the Markov chain search problem and demonstrate how to use it to solve the element distinctness problem. For simplicity, we use the transition matrix P to refer to a Markov chain, where $P = (p_{xy})_{x,y\in X}$ for X being the state space of P and p_{xy} being

the transition probability from x to y. An irreducible and ergodic Markov chain has a unique stationary distribution π, which is also the unique eigenvector of P with eigenvalue 1. Let $M \subseteq X$ be a set of marked elements. In the Markov chain search problem, the objective is to find an $x \in M$. We can perform the following actions: setup, sampling from the π with cost S; update, making a transition with cost U, and checking whether the current state is marked or not with cost C. To solve the search problem classically, we perform a random walk as follows. Start from a point sampled from π and check if it is marked. If not, make a number of transitions on P until it mixes, and then check again. We then repeat this process until a marked state is found. The cost of this random walk algorithm is $O(\mathsf{S} + \frac{1}{\lambda}(\frac{1}{\delta}\mathsf{U} + \mathsf{C}))$, where $\lambda := |M|/|X|$ and δ is the spectral gap of P.

Quantum analogues of random walks, namely, quantum walks, have been developed for solving different problems. In 2003, Ambainis [5] proposed a quantum walk algorithm for solving the element distinctness problem. His algorithm also solves the Markov chain search problem on the Johnson graph with cost $O(\mathsf{S} + \frac{1}{\sqrt{\lambda}}(\frac{1}{\sqrt{\delta}}\mathsf{U} + \mathsf{C}))$. In 2004, Szegedy [45] gave a quantum walk algorithm for more generalized Markov chains with cost $O(\mathsf{S} + \frac{1}{\sqrt{\lambda\delta}}(\mathsf{U} + \mathsf{C}))$. We can view Szegedy's quantum walk as a quantum counterpart of a random walk, where one checks the state after each transition. Szegedy's quantum walk only detects the presence of a marked state, but cannot find one without extra costs. In 2006, Magniez et al. [33] proposed a quantum walk search framework that unified the advantages of the quantum walks in [5] and [45]. In this quantum walk framework, we can perform the following operations:

- **Setup:** with cost S. preparing the initial state $|\pi\rangle = \frac{1}{\sqrt{|X|}} \sum_x \sqrt{\pi_x} |x\rangle$.
- **Update:** with cost U. applying the transformation $|x\rangle |0\rangle \mapsto |x\rangle \sum_{y \in X} \sqrt{p_{xy}} |y\rangle$.
- **Checking:** with cost C, applying the transformation: $|x\rangle \mapsto \begin{cases} -|x\rangle & \text{if } x \in M \\ |x\rangle & \text{otherwise.} \end{cases}$

The main result of [33] is summarized as follows.

Lemma 22 ([33])**.** *Let P be an irreducible and ergodic Markov chain P on X. Let $M \subseteq X$ be a subset of marked elements. Let $\lambda := |M|/|X|$ and δ be the spectral gap of P. Then, there exists a quantum algorithm that with high probability, determines M is empty or finds an $x \in M$ with cost $O(\mathsf{S} + \frac{1}{\sqrt{\lambda}}(\frac{1}{\sqrt{\delta}}\mathsf{U} + \mathsf{C}))$.*

To solve the element distinctness problem, we define a Markov chain, following the work [5, 11, 28]. The state space X is all subsets of $[n]$ with size r. The Markov chain is based on the Johnson graph on X, where an edge is connecting S and S' if and only if $|S \cap S'| = r - 1$. The transition probability on each edge is hence $\frac{1}{r(n-r)}$. A state S is marked when there exist distinct $i, j \in S$ such the ith and the jth items are the same. The Markov chain has spectral gap $\delta \geq 1/r$ (see [28]) and it is easy to verify that $\lambda \geq \binom{n-2}{r-2}/\binom{n}{r} = O(r^2/n^2)$. If we only consider the query complexity, the setup procedure costs r queries, the update procedure costs one query, and the checking procedure does not cost any query. Choosing $r = n^{2/3}$ yields the optimal query complexity $O(n^{2/3})$.

3 Quantum fine-grained complexity

In this section, we give the formal definitions of the quantum fine-grained reduction and quantum strong exponential time hypothesis (QSETH). Moreover, we show that under QSETH, for $d = \text{polylog}(n)$, the lower bounds for $\mathsf{CP}_{n,d}$ and $\mathsf{OV}_{n,d}$ are $n^{1-o(1)}$, which nearly matches the upper bounds given in Theorem 15.

3.1 Quantum fine-grained reduction and QSETH

QSETH is defined based on the assumption that the best quantum algorithm for CNF-SAT is Grover search when the clause width k is large enough.

Hypothesis 23 (QSETH)**.** *For every $\epsilon > 0$, there exists a $k = k(\epsilon) \in \mathbb{N}$ such that no quantum algorithm can solve k-SAT (i.e., satisfiability on a CNF of width k) in $O(2^{(1/2-\epsilon)n})$ time where n is the number of variables. Moreover, this holds even when the number of clauses is at most $c(\epsilon)n$ where $c(\epsilon)$ denotes a constant that depends only on ϵ.*

Obviously, the Grover search can solve CNF-SAT in $\widetilde{O}(2^{n/2})$. To the best of the our knowledge, there is no quantum algorithm that can do better than $O(2^{n/2})$ for any k.

We recall that in the quantum query model, the input of a problem is given by a quantum oracle. Specifically, let P be a problem, and X be an instance of P in the classical setting. Then, in the quantum query model, X will be given by an oracle $\mathcal{O}_X$. We will denote an algorithm or an oracle $\mathcal{A}$ with access to $\mathcal{O}_X$ by $\mathcal{A}(\mathcal{O}_X)$.

We say $\mathcal{A}_\epsilon$ is an ϵ-oracle for problem P, if for every instance $\mathcal{O}_X$, it holds that

$$\Pr[\mathcal{A}_\epsilon(\mathcal{O}_X) = \mathsf{P}(X)] \geq 1 - \epsilon, \tag{8}$$

and the running time is $O(1)$, where $\mathsf{P}(X)$ is the answer of X for problem P.

Definition 24 (Quantum oracles)**.** *Let $X := \{x_1, \ldots, x_n\}$ be an instance of some problem and $\mathcal{O}_X$ be the corresponding quantum oracle. To realize $\mathcal{O}_X$, we do not need to write down the whole X; instead, we can just design a quantum circuit to realize the mapping*

$$|i\rangle |0\rangle \xrightarrow{\mathcal{O}_X} |i\rangle |x_i\rangle . \tag{9}$$

Definition 25 (Quantum fine-grained reduction)**.** *Let $p(n)$ and $q(n)$ be non-decreasing functions of n. Let P and Q be two problems in the quantum query model and $\mathcal{A}_\epsilon$ be an ϵ-oracle for Q with error probability $\epsilon \leq 1/3$. P is quantum (p, q)-reducible to Q, denoted as $(\mathsf{P}, p) \leq_{\text{QFG}} (\mathsf{Q}, q)$, if for every ϵ, there exits a $\delta > 0$, and algorithm R with access to $\mathcal{A}_\epsilon$, a constant d, and an integer $k(n)$, such that for every $n \geq 1$, the algorithm R takes any instance of P of size n and satisfies the following:*

- *R can solve P with success probability at least $2/3$ in time at most $d \cdot p(n)^{1-\delta}$.*

- *R performs at most $k(n)$ quantum queries to A_ϵ. Specifically, in the jth query, let $\mathbf{X}_j := \{X_{1,j}, X_{2,j}, \dots\}$ be a set instances of Q. Then, R realizes the oracles $\{\mathcal{O}_{X_{1,j}}, \mathcal{O}_{X_{2,j}}, \dots\}$ in superposition and applies A_ϵ to solve the instances.*
- *The following inequality holds.*

$$\sum_{j=1}^{k(n)} c(\mathbf{X}_j) \cdot q(n_j)^{1-\epsilon} \leq d \cdot p(n)^{1-\delta},$$

where $c(\mathbf{X}_j)$ is the time required for R to realize the oracles $\{\mathcal{O}_{X_{1,j}}, \mathcal{O}_{X_{2,j}}, \dots\}$ in superposition and $n_j := \max_i |X_{i,j}|$.

In Theorem 25, the input of A_ϵ is given as a quantum oracle such that A_ϵ can be a quantum query algorithm with running time strictly less than the input size. Moreover, the quantum reduction R can realize quantum oracles $\{\mathcal{O}_{X_{1,j}}, \mathcal{O}_{X_{2,j}}, \dots\}$ in superposition, and thus the time required is $\max_i c(X_{i,j})$ (where $c(X_{i,j})$ is the time required to realize $\mathcal{O}_{X_{i,j}}$) instead of $\sum_i c(X_{i,j})$. This also allows R to use fast quantum algorithms to process the information of $A'_\epsilon s$ output (e.g., amplitude amplification).

3.2 Lower bounds for CP, OV, and BCP in higher dimensions under QSETH

Here, we give nearly linear lower bounds for OV and CP under QSETH by showing that there exist quantum fine-grained reductions from SAT to these problems.

Theorem 26. *Assuming QSETH, for all $\epsilon > 0$, there exists a c such that $\mathsf{OV}_{n,c\log n}$ and $\mathsf{CP}_{n,(\log n)^c}$ cannot be solved by any quantum algorithm in time $O(n^{1-\epsilon})$.*

We prove Theorem 26 by showing that there exist quantum fine-grained reductions from CNF-SAT to OV, OV to BCP, and BCP to CP with desired parameters. We first give the reduction from CNF-SAT to OV as a warm-up.

Lemma 27.

$$(\mathsf{CNF\text{-}SAT}_n, 2^{n/2}) \leq_{\mathrm{QFG}} (\mathsf{OV}_{n_1,d_1}, n_1), \tag{10}$$

where $n_1 = 2^{n/2}$ and $d_1 = \Theta(n)$.

Proof. Let ϕ be a CNF formula with n variables and $m = \Theta(n)$ clauses. Let $\mathcal{A}$ be an algorithm for OV. We first recall the classical reduction. Let $\phi := \phi_1 \wedge \cdots \wedge \phi_m$. We divide the n variables into two sets A and B with $|A| = |B| = \frac{n}{2}$. Let $A := \{x_1, \dots, x_{n/2}\}$ and $B := \{x_{n/2+1}, \dots, x_n\}$. We let $S_A := \{a_1, \dots, a_{2^{n/2}}\}$ be all assignments to A and $S_B := \{b_1, \dots, b_{2^{n/2}}\}$ be all assignments to B. We describe two mappings $f_A : S_A \to \{0,1\}^m$ and $f_B : S_B \to \{0,1\}^m$ as follows:

$$f_A(a_i) = [\phi_1(a_i), \dots, \phi_m(a_i)]^T, \text{ and} \tag{11}$$

$$f_B(b_i) = [\phi_1(b_i), \dots, \phi_m(b_i)]^T, \tag{12}$$

where $\phi_j(a_i) = 0$ if a_i is a satisfied assignment for ϕ_j, and $\phi_j(a_i) = 1$ otherwise; we define $\phi_i(b_i)$ in the same way. Let $F_A := \{f_A(a_i) : \ i \in [2^{n/2}]\}$ and $F_B := \{f_B(b_i) : \ i \in [2^{n/2}]\}$. Then, it is obvious that if there exist $v \in F_A$ and $u \in F_B$ such that $\langle v, u\rangle = 0$, then ϕ is satisfiable. However, at first glance, this reduction with $O(2^{n/2})$ running time is not fine-grained since we require the cost of the reduction to be at most $2^{n(1-\delta)/2}$ for some $\delta > 0$ by Theorem 25, but writing down elements in F_A and F_B already takes $\Omega(2^{n/2})$.

Nevertheless, as in Theorem 24, a quantum fine-grained reduction only needs to realize the functions f_A and f_B, which takes $O(mkn)$ time where k is the width of clauses. This is much less than $O(2^{n(1-\delta)/2})$. More specifically, f_A and f_B are oracles for F_A and F_B, and for any quantum query to elements in F_A or F_B, the reduction can implement oracles f_A and f_B:

$$|e, x\rangle |0\rangle \xrightarrow{f_e} |e, x\rangle |f_e(x)\rangle , \tag{13}$$

where $e \in \{A, B\}$, and the time $c(f_e)$ for the reduction to implement f_e for one quantum query is at most $O(kmn)$. Finally, this reduction only uses one oracle (F_A, F_B). If there is an algorithm for OV which succeeds with probability $2/3$, we can boost the success probability of the reduction by repetition. Therefore, $(\mathsf{CNF\text{-}SAT}, 2^{n/2})$ is quantum reducible to $(\mathsf{OV}_{n_1,d_1}, n_1)$. □

Then, to prove $(\mathsf{CNF\text{-}SAT}, 2^{n/2}) \leq_{\mathrm{QFG}} (\mathsf{CP}_{n_3,d_3}, n_3)$, we show that $(\mathsf{BCP}_{n_2,d_2}, n_2) \leq_{\mathrm{QFG}} (\mathsf{CP}_{n_3,d_3}, n_3)$ and $(\mathsf{OV}_{n_1,d_1}, n_1) \leq_{\mathrm{QFG}} (\mathsf{BCP}_{n_2,d_2}, n_2)$, where n_2, n_3, d_2, d_3 are some functions of n specified in the following lemmas.

Lemma 28. *For $d = \Theta(\log n)$,*

$$(\mathsf{BCP}_{n,d}, n) \leq_{\mathrm{QFG}} (\mathsf{CP}_{n',d'}, n'), \tag{14}$$

where $n' = n^{O(1)}$ and $d' = (\log n)^c$ for some constant c and all points have $\{0, 1\}$ entries with the Hamming metric.

Remark 29. *The points have coordinate entries in $\{0, 1\}$, and the Hamming metric is equivalent to distance in ℓ_2-metric (up to power of 2) in this case. Therefore, in the proof of Lemma 28, we can consider the Hamming distance between points instead of ℓ_2 distance without loss of generality.*

We first introduce the classical reductions in [29] and some results we will use to prove Lemma 28.

Classical reduction We can consider an instance of BCP with two sets of points A and B as a weighted complete bipartite graph $K_{n,n}$, where the vertices are the points in these two sets and edges' weights are equal to the distances between the corresponding points. Then, solving BCP is equivalent to find an edge with the minimum weight in this graph. However, we cannot directly apply the algorithm for CP on this graph since there could be two points in the same set (no edge connecting them) that have a smaller distance than any pairs of points in two sets (connected by an edge). To overcome this difficulty, we can "stretch" the points to make the points in the same set far from each other, which is characterized by the contact dimension of a graph:

Definition 30 (Contact Dimension)**.** *For any graph $G = (V, E)$, a mapping $\tau : V \to \mathbb{R}^d$ is said to realize G if for some $\beta > 0$, the following holds for every distinct vertices u, v:*

$$\begin{aligned} &\|\tau(u) - \tau(v)\|_2 = \beta \text{ if } \{u, v\} \in E, \\ &\|\tau(u) - \tau(v)\|_2 > \beta \text{ otherwise.} \end{aligned} \tag{15}$$

The contact dimension of G, denoted by $\mathrm{cd}(G)$, is the minimum $d \in \mathbb{N}$ such that there exists $\tau : V \to \mathbb{R}^d$ realizing G.

That is, with the help of τ, we can restrict the optimal solution of CP to be the points connected by an edge in G. But we cannot realize the whole complete bipartite graph since $\mathrm{cd}(K_{n,n}) = \Theta(n)$, which makes the dimension of the CP instance too large. [29] showed that we can realize a subgraph of $K_{n,n}$ and apply permutations to its vertices such that the union of these subgraphs cover $K_{n,n}$. In this way, BCP can be computed by solving CP on each subgraph and outputting the best solution. More specifically, the reduction in [29] relies on the following theorem:

Theorem 31 (Theorem 4.2 in [29])**.** *For every $0 < \delta < 1$, there exists a log-dense sequence $(n_i)_{i\in\mathbb{N}}$ such that, for every $i \in \mathbb{N}$, there is a bipartite graph $G_i = (A_i \dot\cup B_i, E_i)$ where $|A_i| = |B_i| = n_i$ and $|E_i| \geq \Omega(n_i^{2-\delta})$, such that $\mathrm{cd}(G_i) = (\log n_i)^{O(1/\delta)}$. Moreover, for all $i \in \mathbb{N}$, a realization $\tau : A_i \dot\cup B_i \to \{0,1\}^{(\log n_i)^{O(1/\delta)}}$ of G_i can be constructed in $n_i^{2+o(1)}$ time.*

The log-dense sequence is defined as follows:

Definition 32. *A sequence $(n_i)_{i\in\mathbb{N}}$ of increasing positive integers is log-dense if there exists a constant $c \geq 1$ such that $\log n_{i+1} \leq c \cdot \log n_i$ for all $i \in \mathbb{N}$.*

They also showed that, the permutations for covering the complete bipartite graph can be efficiently found, as shown in the following lemma.

Lemma 33 (Lemma 3.11 in [29])**.** *For any bipartite graph $G(A \dot\cup B, E_G)$ where $|A| = |B| = n$ and $E_G \neq \emptyset$, there exist side-preserving permutations $\pi_1, \ldots, \pi_k : A \cup B \to A \cup B$ where $k \leq \frac{2n^2 \ln n}{|E_G|} + 1$ such that*

$$\bigcup_{i\in[k]} E_{G_{\pi_i}} = E_{K_{n,n}}. \tag{16}$$

Moreover, such permutations can be found in $O(n^6 \log n)$ time.

Now, we are ready to state the quantum fine-grained reduction by "quantizing" the classical reduction.

Proof of Theorem 28. Let A, B be the two sets of input points of BCP. Suppose for BCP, there is an input oracle $\mathcal{O}_{\mathsf{BCP}}$ which, given an index, returns the corresponding point:

$$|b\rangle\,|i\rangle\,|0\rangle \xrightarrow{\mathcal{O}_{\mathsf{BCP}}} \begin{cases} |b\rangle\,|i\rangle\,|x_i\rangle & \text{if } b = 0, \\ |b\rangle\,|i\rangle\,|y_i\rangle & \text{if } b = 1, \end{cases} \tag{17}$$

where x_i is the i-th point in the set A and y_i is the i-th point in the set B. The sizes of A and B are both equal to n and each point is in $\{0,1\}^{d_1}$, where $d_1 = \Theta(\log n)$ is the dimension of BCP.

For CP, suppose there is a quantum algorithm $\mathcal{A}$ such that for m points in $\{0,1\}^{d_2}$ given by an oracle $\mathcal{M}_{CP}$, $\mathcal{A}^{\mathcal{M}_{CP}}$ returns the closest pair of these n points with probability at least $2/3$.

Then we need to transform $\mathcal{O}_{\mathsf{BCP}}$ to some oracles $\mathcal{M}_i$ for CP, such that by running $\mathcal{A}$ with $\mathcal{M}_i$ as input oracles, we can get the bichromatic closest pair between A and B. The reduction has four steps:

1. Pre-processing. We first follow the classical reduction to pre-process the input points of BCP. For some integer $n' \leq n^{0.1}$, we can partition A and B into n'-size subsets:

$$\begin{aligned} A &= A_1 \dot{\cup} \cdots \dot{\cup} A_r, \\ B &= B_1 \dot{\cup} \cdots \dot{\cup} B_r, \end{aligned} \tag{18}$$

where $r = \lfloor n/n' \rfloor$. Here, we assume that n is divisible by n'. It follows that

$$\mathsf{BCP}(A,B) = \min_{i,j\in[r]} \mathsf{BCP}(A_i, B_j). \tag{19}$$

Then, we use the algorithm in [29] to construct k mappings $f_1, \ldots, f_k : [2n'] \to \{0,1\}^{d'}$ such that

$$\mathsf{BCP}(A_i, B_j) = \min_{t\in[k]} \mathsf{CP}(f_t(A_i) \cup f_t(B_j)) \quad \forall i,j \in [\lfloor n/n' \rfloor]. \tag{20}$$

More specifically, we pick n' to be the largest number in a log-dense sequence that is smaller than $n^{0.1}$. Then, we apply Theorem 31 to classically construct a bipartite graph $G(A \cup B, E)$ with n' vertices in each side and a realization τ. By choosing $\delta = \epsilon/2$ in Theorem 31, the graph G has $|E| = \Omega(n'^{2-\epsilon/2})$ edges. And we can get $2n'$ 0/1-strings of length $(\log n')^{O(2/\epsilon)}$:

$$\tau_i^A = \tau(u_i) \quad \forall u_i \in A, \qquad \text{and} \qquad \tau_i^B = \tau(v_i) \quad \forall v_i \in B. \tag{21}$$

In order to cover the complete bipartite graph, we run the classical algorithm (Theorem 33) to find k permutations $\pi_1, \ldots, \pi_k : [n'] \to [n']$, where k is a parameter to be chosen later.

Then, we can define the mappings as follows:

$$f_t(u) = \begin{cases} x_v \circ \left(\tau^A_{\pi_t(w)}\right)^{d+1} & \text{if } 1 \leq u \leq n' \\ y_v \circ \left(\tau^B_{\pi_t(w)}\right)^{d+1} & \text{if } n' < u \leq 2n' \end{cases} \quad \forall t \in [k], u \in [2n'], \tag{22}$$

where $\circ$ means string concatenation and $(s)^{d+1}$ denotes $d+1$ copies of the string s. For a point $p \in A_i \cup B_j$, $u \in [2n']$ is the index in this union-set, $v \in [n]$ is the index in the ground set A or B, and $w \in [n']$ is the index in the subset A_i or B_j. Further, if $1 \leq u \leq n'$, then $w := u$; otherwise, $w := u - n'$.

2. Oracle construction. For $i, j \in [r], t \in [k]$, we then construct the input oracle $\mathcal{M}_{i,j,t}$ for the problem $\mathsf{CP}(f_t(A_i) \cup f_t(B_j))$. For a query index $u \in [2n']$,

$$M_{i,j,t} |u\rangle |0\rangle = |u\rangle |f_t(u)\rangle . \tag{23}$$

With the help of the input oracle $\mathcal{O}_{\mathsf{BCP}}$, we can implement $\mathcal{M}_{i,j,t}$ in the following way:

1. Prepare an ancilla qubit $|b\rangle$ such that $b = 1$ if $u > n'$.
2. Transform $|u\rangle$ to $|v\rangle$, the index of the point in A or B, based on the value of b. Note that the index is unique. Hence, this transformation is unitary and can be easily achieved by a small quantum circuit.
3. Query $\mathcal{O}_{\mathsf{BCP}}$ with input $|b\rangle |v\rangle$. Assume $b = 0$. Then,
$$|b\rangle |v\rangle |0\rangle \xmapsto{\mathcal{O}_{\mathsf{BCP}}} |b\rangle |v\rangle |x_v\rangle . \tag{24}$$
4. Similar to the second step, the index w of the point in A_i and B_j can be computed from v by a unitary transformation:
$$|b\rangle |v\rangle |x_v\rangle \mapsto |b\rangle |w\rangle |x_v\rangle \tag{25}$$
5. Since each w corresponds to a unique string $\tau^A_{\pi_t(w)}$, we can attach $d+1$ copies of this string to the remaining quantum registers:
$$|b\rangle |w\rangle |x_v\rangle \mapsto |b\rangle |w\rangle |x_v\rangle \left| \left(\tau^A_{\pi_t(w)}\right)^{d+1} \right\rangle . \tag{26}$$
6. By recovering u from w, we get the final state:
$$|u\rangle |f_t(u)\rangle = |u\rangle \left| x_v, \left(\tau^A_{\pi_t(w)}\right)^{d+1} \right\rangle . \tag{27}$$

3. Query process By Eqs. (19) and (20), we have

$$\mathsf{BCP}(A, B) = \min_{i,j\in[r],t\in[k]} \mathsf{CP}(f_t(A_i) \cup f_t(B_j)). \tag{28}$$

Hence, we can use quantum minimum-finding algorithm in Theorem 15 over the sub-problems to find the minimum solution. For each sub-problem, we can run the algorithm for CP with $\mathcal{M}_{i,j,t}$ as the input oracle.

4. Post-processing. In case that n is not divisible by n', let the remaining points in A and B be A_{res}, B_{res}, respectively. Then, we can use Grover search to find the closest pair between A_{res} and B, and between B_{res} and A. Then, compare the answer to the previously computed result and pick the smaller one.

Correctness. In this reduction, we do not change the constructions of the mappings $\{f_i\}_{i\in[k]}$. By [29], Eq. (28) is correct in the classical setting. Hence, it also holds in the quantum setting, and we can use Grover search to find the minimum solution. However, since the algorithm $\mathcal{A}$ for CP has success probability 2/3, for each tuple $(i, j, t) \in [r] \times [r] \times [k]$, we need to run $\mathcal{A}^{\mathcal{M}_{i,j,t}}$ $O(\log n)$ times to boost the success probability to at least $1 - \frac{1}{n}$. Then, by the union bound, the probability that all queries in the Grover search are correct is at least 99/100. Hence, by Theorem 9, the overall success probability is at least 2/3.

Running Time of the Reduction. The running time of the pre-processing step consists of two parts: (1) constructing the graph G and its realization τ; (2) finding k permutations. For the first part, by Theorem 31, it can be done in $n'^{2+o(1)}$ time. For the second part, we pick $k = O(\frac{2n'^2 \log n'}{n'^{2-\epsilon/2}}) = O(n'^{\epsilon/2} \log n')$, and by Theorem 33, it can be done in $O(n'^6 \log n')$ time. Hence, the total running time of pre-processing step is $n'^{2+o(1)} + O(n'^6 \log n') = \widetilde{O}(n^{0.6})$.

The oracle construction can be done "on-the-fly". More specifically, given the strings $\{\tau_i^A, \tau_i^B\}_{i\in[n']}$, and permutations $\{\pi_i\}_{i\in[k]}$, for each query index u, we can simulate the oracle $\mathcal{M}_{i,j,t}$ defined in Eq. (23) in $c(\mathcal{M}_{i,j,t}) = O(d_2) = (\log n')^{\Omega(1)} = \widetilde{O}(1)$ time.

In the query process, for each CP instance indexed by (i, j, t), suppose $\mathcal{A}^{\mathcal{M}_{i,j,t}}$ gets the answer in time $q(n') = n'$. Moreover, for each time $\mathcal{A}$ querying the input oracle $\mathcal{M}_{i,j,t}$, we need to spend $c(\mathcal{M}_{i,j,t})$ time to simulate the oracle. And we also have $O(\log n)$ runs for each instance. Hence, the total running time for each CP is at most

$$n'^{1-\epsilon} \cdot \widetilde{O}(1) \cdot O(\log n) = \widetilde{O}(n'^{1-\epsilon}). \tag{29}$$

Then, we use Grover's search algorithm over $r^2 \cdot k$ instances, which can be done by querying $\widetilde{O}(\sqrt{r^2 \cdot k})$ instances by Theorem 9. Therefore, for any $\epsilon > 0$, we have

$$\widetilde{O}(\sqrt{r^2 k}) \cdot q(n')^{1-\epsilon} \cdot c(\mathcal{M}_{i,j,t}) \cdot O(\log n) = \widetilde{O}(\sqrt{(n/n')^2 k} \cdot (n')^{1-\epsilon}) \tag{30}$$

$$\leq \widetilde{O}(n \cdot (n')^{-\epsilon}) \leq \widetilde{O}(n \cdot n^{-\epsilon/2}) \leq n^{1-\delta}, \tag{31}$$

where the first inequality follows from $k = O(n'^{\epsilon/2} \log n')$ as shown in [29] and the last inequality follows by setting $\delta = \epsilon/10$.

For the post-processing step, the sizes of A_{res} and B_{res} are at most n'. The running time is

$$O(\sqrt{n \cdot n'} \cdot \log n) \leq \widetilde{O}(n^{0.55}). \tag{32}$$

Therefore, for any $\epsilon > 0$, there exists a $\delta > 0$ such that the Eq. (30) holds and the total reduction time is $O(n^{1-\delta})$. By Theorem 25, BCP_{n,d_1} can be quantum fine-grained reduced to CP_{n,d_2}. This completes the proof of this lemma. □

Finally, we show that $(\mathsf{OV}_{n,d}, n) \leq_{\mathrm{QFG}} (\mathsf{BCP}_{n,d'}, n)$ by quantizing the reduction in [29] following the same idea.

Lemma 34. *For $d = \Theta(\log n)$,*

$$(\mathsf{OV}_{n,d}, n) \leq_{\mathrm{QFG}} (\mathsf{BCP}_{n,d'}, n), \tag{33}$$

where $d' = \Theta(\log n)$.

Proof. For an OV instance with sets of vectors A and B, let $\mathcal{O}_{\mathsf{OV}}$ be the input oracle such that:

$$\mathcal{O}_{\mathsf{OV}} |i\rangle |0\rangle = \begin{cases} |i\rangle |a_i\rangle & \text{if } i \in A, \\ |i\rangle |b_i\rangle & \text{if } i \in B. \end{cases} \tag{34}$$

where $a_i, b_i \in \{0,1\}^d$.

Then, similar to the classical reduction, we can construct mappings $f_A, f_B : \{0,1\}^d \to \{0,1\}^{5d}$ such that

$$f_A(a_i)_{5j-4:5j} = \begin{cases} 11000 & \text{if } a_i(j) = 0 \\ 00110 & \text{if } a_i(j) = 1 \end{cases} \quad \forall j \in [d], \tag{35}$$

and

$$f_B(b_i)_{5j-4:5j} = \begin{cases} 10100 & \text{if } b_i(j) = 0, \\ 01001 & \text{if } b_i(j) = 1. \end{cases} \quad \forall j \in [d]. \tag{36}$$

By the classical reduction, we have

$$\mathsf{OV}(A,B) = 1 \text{ if and only if } \mathsf{BCP}(f_A(A), f_B(B)) = 2d \tag{37}$$

under Hamming distance.

Also, note that we can simulate the input oracle $\mathcal{O}_{\mathsf{BCP}}$ by first querying the oracle $\mathcal{O}_{\mathsf{OV}}$ to get the vector, then applying the corresponding mapping f_A or f_B, which can be done in $c(\mathcal{O}_{\mathsf{BCP}}) = O(d)$ time. Let the running time of the algorithm for BCP be $q(n) = n$. Then for any $\epsilon > 0$,

$$q(n)^{1-\epsilon} \cdot c(\mathcal{O}_{\mathsf{BCP}}) = n^{1-\epsilon} \cdot \Theta(\log n) \leq n^{1-\delta} \tag{38}$$

for some small $\delta > 0$. Hence, by Theorem 25, $(\mathsf{OV}_{n,d}, n) \leq_{\mathrm{QFG}} (\mathsf{BCP}_{n,d'}, n)$. □

Proof of Theorem 26. We can prove the theorem by contradiction following Theorem 27, Theorem 34, and Theorem 28. Specifically, suppose that there exists an $\epsilon > 0$, for all $d = \Theta(\log n)$, there exists a quantum algorithm which can solve OV in time $O(n^{1-\epsilon})$. Then, we can obtain a quantum algorithm for $\mathsf{CNF\text{-}SAT}$, which runs in time $O(2^{n/2(1-\epsilon)})$ by Theorem 27. This contradicts QSETH. The proof for CP is the same. □

3.3 Quantum lower bound for BCP in nearly-constant dimensions under QSETH

A byproduct of the previous subsection is a quantum lower bound for BCP in higher dimensions (i.e., $d = \mathrm{polylog}(n)$) under QSETH (Theorem 34). In this subsection, we show that this quantum lower bound for BCP even holds for nearly-constant dimensions (i.e., $d = c^{\log^*(n)}$). The main result of this subsection is the following theorem.

Theorem 35. *Assuming QSETH, there is a constant c such that BCP in $c^{\log^*(n)}$ dimensions requires $n^{1-o(1)}$ time for any quantum algorithm.*

We will "quantize" the results by Chen [17] to prove this theorem. More specifically, we first show a quantum fine-grained self-reduction of OV from $\log n$ dimensions with binary entries to $2^{\log^*(n)}$ dimensions with integer entries ($\mathbb{Z}$-OV). Then, we give a quantum fine-grained reduction from $\mathbb{Z}$-OV to BCP in nearly-constant dimensions.

Definition 36 (Integral Orthogonal Vector, $\mathbb{Z}$-OV)**.** *Given two sets A, B of n vectors in $\mathbb{Z}^d$, find a pair of vectors $a \in A$ and $b \in B$ such that $\langle a, b\rangle = 0$, where the inner product is taken in $\mathbb{Z}$.*

We use $\mathbb{Z}\text{-}\mathsf{OV}_{n,d}$ to denote $\mathbb{Z}\text{-}\mathsf{OV}$ with n vectors of d dimension in each set. We then recap a theorem in [17]:

Theorem 37 ([17, Theorem 4.1])**.** *Let b, ℓ be two sufficiently large integers. There is a classical reduction $\psi_{b,\ell} : \{0,1\}^{b\cdot\ell} \to \mathbb{Z}^{\ell}$ and a set $V_{b,\ell} \subseteq \mathbb{Z}$, such that for every $x, y \in \{0,1\}^{b\cdot\ell}$,*

$$\langle x, y\rangle = 0 \;\Leftrightarrow\; \langle \psi_{b,\ell}(x), \psi_{b,\ell}(y)\rangle \in V_{b,\ell} \tag{39}$$

and

$$0 \le \psi_{b,\ell}(x)_i < \ell^{6^{\log^*(b)}\cdot b} \tag{40}$$

for all possible x and $i \in [\ell]$. Moreover, the computation of $\psi_{b,\ell}(x)$ takes $\mathrm{poly}(b\cdot\ell)$ time, and the set $V_{b,\ell}$ can be constructed in $O\left(\ell^{O(6^{\log^*(b)}\cdot b)} \cdot \mathrm{poly}(b\cdot\ell)\right)$ time.

Note that the size of $V_{b,\ell}$ is at most $\ell^{2\cdot 6^{\log^*(b)}\cdot b+1}$. The following lemma gives a quantum fine-grained reduction from OV to $\mathbb{Z}\text{-}\mathsf{OV}$:

Lemma 38. *For $d = \Theta(\log n)$,*

$$(\mathsf{OV}_{n,d}, n) \le_{\mathrm{QFG}} (\mathbb{Z}\text{-}\mathsf{OV}_{n,d'}, n). \tag{41}$$

where $d' = 2^{O(\log^* n_2)}$.

Proof. Consider an $\mathsf{OV}_{n,d}$ with $d = c\cdot\log n$, where c is an arbitrary constant. We choose $\ell := 7^{\log^* n}$ and $b := d/\ell$. Then, we can apply Theorem 37 to get the mapping function $\psi_{b,\ell}$ and the set $V_{b,\ell}$. For each $v \in V_{b,\ell}$, we'll construct an instance of $\mathbb{Z}\text{-}\mathsf{OV}_{n,\ell+1}$ as follows:

1. Let $|i\rangle$ be the input query index of $\mathbb{Z}\text{-}\mathsf{OV}_{n,\ell+1}$.
2. Query $\mathsf{OV}_{n,d}$'s input oracle $\mathcal{O}_{\mathsf{OV}}$ and get the vector $|i, x\rangle$.
3. Compute the mapping $\psi_{b,\ell}$ and get $|i, x\rangle\,|\psi_{b,\ell}(x)\rangle$.
4. If $x \in A$, then attach 1 to the end of the register: $|i, x\rangle\,|\psi_{b,\ell}(x), 1\rangle$. If $x \in B$, then attach $-v$ to the end: $|i, x\rangle\,|\psi_{b,\ell}(x), -v\rangle$.
5. Use $\mathcal{O}_{\mathsf{OV}}$ to erase x and return the final input state $|i\rangle\,|\psi_{b,\ell}(x), 1\rangle$ or $|i\rangle\,|\psi_{b,\ell}(x), -v\rangle$.

For each instance, we can use the quantum oracle for $\mathbb{Z}\text{-}\mathsf{OV}_{n,\ell+1}$ to check the orthogonality. $\mathsf{OV}_{n,d}$ is YES if and only if there exists a YES-instance of $\mathbb{Z}\text{-}\mathsf{OV}_{n,\ell+1}$.

Correctness. The correctness follows from Eq. (39):

$$\langle x, y\rangle = 0 \Leftrightarrow \langle \psi_{b,\ell}(x), \psi_{b,\ell}(y)\rangle = v \in V_{b,\ell} \Leftrightarrow \langle [\psi_{b,\ell}(x),\ 1], [\psi_{b,\ell}(y),\ -v]\rangle = 0. \tag{42}$$

Reduction time. Note that for $\ell = 7^{\log^* n}$ and $b = d/\ell$, we have:

$$\log\left(\ell^{O(6^{\log^*(d)} \cdot b)}\right) = \log \ell \cdot O\left(6^{\log^*(d)} \cdot (d/\ell)\right) \tag{43}$$

$$= O\left(\log^*(n) \cdot 6^{\log^* n} \cdot c \log n / 7^{\log^* n}\right) \tag{44}$$

$$= o(\log n). \tag{45}$$

This implies that $|V_{b,\ell}| \leq \ell^{2 \cdot 6^{\log^*(b)} \cdot b + 1} \leq n^{o(1)}$. Hence, the number of $\mathbb{Z}$-$\mathsf{OV}_{n,\ell+1}$ instances is $n^{o(1)}$ and the running time for compute $V_{b,\ell}$ is $n^{o(1)}$. And for each input query, the oracle for $\mathbb{Z}$-$\mathsf{OV}_{n,\ell+1}$ can be simulated in $c(\mathcal{O}_{\mathbb{Z}\text{-}\mathsf{OV}}) = \mathrm{poly}(d) = \mathrm{poly}(\log n)$ time. We can show that for every $\epsilon > 0$, if $\mathbb{Z}$-$\mathsf{OV}_{n,\ell+1}$ can be decided in $n^{1-\epsilon}$ time, then

$$\sum_{v \in V_{b,\ell}} n^{1-\epsilon} \cdot c(\mathcal{O}_{\mathbb{Z}\text{-}\mathsf{OV}}) = n^{o(1)} \cdot n^{1-\epsilon} \cdot \mathrm{poly}(\log n) \leq n^{1-\delta} \tag{46}$$

for some $\delta > 0$, which satisfies the definition of quantum fine-grained reduction (Theorem 25).

Therefore, $\mathsf{OV}_{n,O(\log n)}$ is quantum fine-grained reducible to $\mathbb{Z}$-$\mathsf{OV}_{n,2^{O(\log^*(n))}}$. □

Then, we give a quantum fine-grained reduction from $\mathbb{Z}$-OV to BCP:

Lemma 39. *For $d = 2^{O(\log^* n)}$,*

$$(\mathbb{Z}\text{-}\mathsf{OV}_{n,d}, n) \leq_{\mathrm{QFG}} (\mathsf{BCP}_{n,d'}). \tag{47}$$

where $d' = d^2 + 2$.

Proof. We remark here that this proof closely follows that for Theorem 4.3 in [17]. We nonetheless give it here as some details are different.

For an $\mathbb{Z}$-$\mathsf{OV}_{n,d}$ instance with $(k \cdot \log n)$-bit entries, we construct a BCP instance as follows:

1. For $x \in A$, construct a vector $x' \in \mathbb{Z}^{d^2}$ such that $x'_{i,j} = x_i \cdot x_j$. Here, we index a d^2-dimensional vector by $[d] \times [d]$. Similarly, for $y \in B$, construct a vector $y' \in \mathbb{Z}^{d^2}$ such that $y'_{i,j} = -y_i \cdot y_j$.

2. Choose $W := (d^2 + 1) \cdot n^{4k}$. For each x', construct a vector $x'' \in \mathbb{R}^{d^2+2}$ such that

$$x'' = \left[x',\ \sqrt{W - \|x'\|_2^2},\ 0\right]. \tag{48}$$

   For each y', construct a vector $y'' \in \mathbb{R}^{d^2+2}$ such that

$$y'' = \left[y',\ 0,\ \sqrt{W - \|y'\|_2^2}\right]. \tag{49}$$

Then, we claim that the $\mathbb{Z}$-OV instance is YES if and only if the BCP instance has the minimum distance $\leq \sqrt{2W}$.

Correctness. First note that $\|x'\|_2^2 \leq d^2 \cdot (2^{k \log n})^4 = d^2 \cdot n^{4k}$. Hence, $W - \|x'\|_2^2 > 0$ and $W - \|y'\|_2^2 > 0$. For any x'' and y'' in the new constructed instance of BCP, we have

$$\begin{aligned} \|x'' - y''\|_2^2 &= \|x''\|_2^2 + \|y''\|_2^2 - 2 \cdot \langle x'', y'' \rangle && (50) \\ &= 2 \cdot W - 2 \cdot \langle x', y' \rangle && (51) \\ &= 2 \cdot W - 2 \cdot \sum_{(i,j) \in [d] \times [d]} x_i \cdot x_j \cdot (-y_j \cdot y_j) && (52) \\ &= 2 \cdot W + 2 \cdot (\langle x, y \rangle)^2. && (53) \end{aligned}$$

Hence,

$$\langle x, y \rangle = 0 \;\Leftrightarrow\; \|x'' - y''\|_2^2 = 2W. \tag{54}$$

Reduction time. We can see from the above description that the input mapping function is simple and can be computed by a small quantum circuit in $O(d^2) = O(2^{O(\log^*(n))})$ time. Hence, we have $c(\mathcal{O}_{\mathsf{BCP}}) = O(2^{O(\log^*(n))})$. Also, by Theorem 25, it's easy to check that this is indeed a quantum fine-grained reduction from $\mathbb{Z}$-OV to BCP. □

Now Theorem 35 follows immediately from Theorem 38 and Theorem 39:

Proof of Theorem 35. Let $\epsilon > 0$ be some constant. Suppose we can solve $\mathsf{BCP}_{n,c^{\log^*(n)}}$ in $n^{1-\epsilon}$ time for all constant $c > 0$. Then, by Theorem 38 and Theorem 39, we can also solve $\mathsf{OV}_{n,c' \log n}$ in $n^{1-\delta}$ time for some $\delta > 0$ and any $c' > 0$. However, this contradicts QSETH by Theorem 26. Therefore, assuming QSETH, there exists a constant c such that $\mathsf{BCP}_{n,c^{\log^*(n)}}$ requires $n^{1-o(1)}$ time. □

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